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11/10/2018

Problem Set 11
More Combinatorics

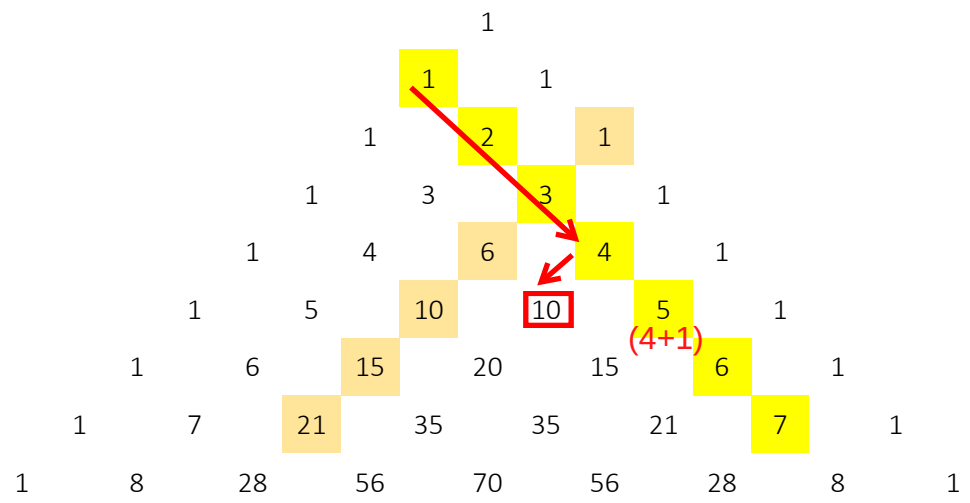
1. Jacob's "Mathematics A Human Endeavor"
    1. Find the values of 11^2, 11^3, 11^4
       $11^2 = 121$
       $11^3 = 1331$
       $11^4 = 14641$
    2. What do these numbers have to do with pascal's triangle?
       If the row of the Pascal's triangle was indexed starting at 0 (as shown below), 11^{index} = the combined values of the row in Pascal's triangle. For example, $11^0 = 1$ and if you look at row index 0 there is a sole 1. Additionally, $11^4 = 14641$ which are the values displayed in row index 4 (see Pascal's triangle below).

index										
0						1				
1					1		1			
2				1		2		1		
3			1		3		3		1	
4		1		4		6		4		1

    3. Now find the value of 11^5. Does it fit the pattern?
       $11^5 = 161051$ - Originally, I though "well poop, the rule is no longer valid" however after taking a peak at mathisfun (https://www.mathsisfun.com/pascals-triangle.html) an amazing trick was shown! The numbers of 1 5 10 10 5 1 are found in Pascal's triangle at row with index 5 however the values converge on each other so the rule remains valid (see table and triangle below).

	Pascal's Values	Resulting Numerical Combination
Start	1 5 10 10 5 1	[]
1 remains, add to list	5 10 10 5 1	[1]
Take sum of next two values (5 + 1), add to list	0 10 5 1	[1 6]
0 eliminated	10 5 1	[1 6]
Place remaining values in list	[]	[1 6 10 5 1] = 11^5 = 161051

index											
0						1					
1					1		1				
2				1		2		1			
3			1		3		3		1		
4		1		4		6		4		1	
5	1		5		10		10		5		1



4. Using the row diagonal row (left to right, highlighted in yellow above) with the values of 1,2,3,4,5,6,7 - What is the sum of:
    a. Its first two numbers – 1 + 2 = 3
    b. Its first three numbers - 1 + 2 + 3 = 6
    c. Its first four numbers - 1 + 2 + 3 + 4 = 10
    d. Its first give numbers - 1 + 2 + 3 + 4 + 5 = 15
    e. Its first six numbers - 1 + 2 + 3 + 4 + 5 + 6 = 21
5. What do you notice about the answers
    a. The sum of any number of elements in the triangle, along that diagonal line, is equal to the value directly to the left of the max element + 1. For example, the sum of the first 4 elements (1+2+3+4) is 10 which can be found directly to the left of 4+1 (follow red arrows).
6. Using the row diagonal row (right to left, highlighted in orange above) with the values of 1,3,6,10,15,21 - What is the sum of:
    a. Its first two numbers - 1 + 3 = 4
    b. Its second and third numbers - 3 + 6 = 9
    c. Its third and fourth numbers - 6 + 10 = 16
    d. Its fourth and fifth numbers - 10 + 15 = 25
    e. Its fifth and sixth numbers - 15 + 21 = 36
7. The sums form a certain number sequence. What is it?
    a. Squares! This sequence starts at 2^2 which = 4, the 3^2 which = 9 and so on and so forth.

8. Sum of sloping rows (image not included). The first three sums are 1,2,3 (shown below), What are the sums of numbers in the next five rows?

   1 = 1
   1 + 1 = 2
   1 + 2 = 3
   1 + 3 + 1 = 4
   1 + 4 + 3 = 8
   1 + 5 + 6 + 1 = 13
   1 + 6 + 10 + 4 = 21
   1 + 7 + 15 + 10 + 1 = 34

9. These eight sums from a certain number sequence. What sequence is it?
   a. Fibonacci's sequence. Each of the values is the sum of the two preceding values.

N	0	1	2	3	4	5	6	7	8
Sum(N)	0	0+1=1	1+1=2	1+2=3	2+3=5	3+5=8	5+8=13	8+13=21	13+21=34

$\boxed{2}$ PROBLEM #2 P.421 Sec. 6.4

$(x+y)^5$

a) Combinatorial Reasoning

$(x+y)(x+y)(x+y)(x+y)(x+y)$

$= xxxxx + xxxxy + xxxyx + xxxyy + xxyxx$

$+ xxyyx + xxyyy + xyxxx + xyyxx$

$+ xyyyx + xyyyy + yxxxx + yyxxx$

$+ yyyxx + yyyyx + yyyyy$

$$\boxed{= x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5}$$

b) Binomial Thm

$$(x+y)^5 = \sum_{j=0}^{5} \binom{5}{j} x^{5-j} y^j$$

$$= \binom{5}{0} x^5 y^0 + \binom{5}{1} x^{5-1} y^1 + \binom{5}{2} x^{5-2} y^2 + \binom{5}{3} x^{5-3} y^3$$

$$+ \binom{5}{4} x^{5-4} y^4 + \binom{5}{5} x^{5-5} y^5$$

$$= \left(\frac{5!}{5!0!}\right) x^5 + \left(\frac{5!}{4!1!}\right) x^4 y + \left(\frac{5!}{3!2!}\right) x^3 y^2 + \left(\frac{5!}{2!3!}\right) x^2 y^3$$

$$+ \left(\frac{5!}{1!4!}\right) x y^4 + \left(\frac{5!}{0!5!}\right) y^5$$

$$\boxed{= 1x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5}$$

3 PROBLEM #9 P 432 Sec. 6.5

a) 8 bagels possible (n)
choose 6 (r)

$C(n+r-1, r) = C(n+r-1, n-1)$

$C(8+6-1, 6) = C(8+6-1, 7)$

$C(13,6) = C(13,7) = \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = \boxed{1716}$

b) 8 bagels possible (n)
choose 12 (r)

$C(n+r-1, r) = C(n+r-1, n-1)$

$C(8+12-1, 12) = C(8+12-1, 8-1)$

$C(19, 12) = C(19, 7)$

$= \frac{19 \cdot 18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7} = \boxed{50388}$

c) 8 bagels possible (n)
choose 24 (r)

$C(n+r-1, r) = C(n+r-1, n-1)$

$C(8+24-1, 24) = C(8+24-1, 8-1)$

$C(31, 24) = C(31, 7)$

$= \frac{31 \cdot 30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7} = \boxed{2629575}$

$\boxed{3 \text{ CONT}}$ PROBLEM #9 P 432 SEC 6.5

d) 8 bagels
choose 12, at least one of each

to have a minimum of one of each kind 8 bagels must be chosen at the start, one of each kind. The remaining 4 (12 - 8) bagels can be of any type. Leaving:

8 bagels (n)
choose 4

$c(n+r-1, r) = c(n+r-1, n-1)$

$c(8+4-1, 4) = c(8+4-1, 8-1)$

$c(11, 4) = c(11, 7)$

$= \frac{11 \cdot 10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3 \cdot 4} = \boxed{330}$

e) 8 bagels, choose 12 $\boxed{\geq 3 \text{ egg}}$ + ≤ 2 salty
starting w/the 3 egg bagels that are required → the remaining 9 yield:

$c(8+9-1, 9) = c(8+9-1, 7)$

$c(16, 9) = c(16, 7)$

$= \frac{16 \; 15 \; 14 \; 13 \; 12 \; 11 \; 10}{1 \; 2 \; 3 \; 4 \; 5 \; 6 \; 7} = 11440$

now we must subtract the possibility of including > 3 salty bagels from the 9 remaining bagels → $c(8+(9-3)-1, (9-3))$

$11440 - c(13, 6) = 11440 - \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = 11440 - 1716$

$\boxed{= 9724}$

$\boxed{4}$ PROBLEM # 32 p 422 Sec 6.4

Prove the binomial thm by <u>induction</u>

<u>binomial thm:</u>

for all $n \in \mathbb{N}$,

$$(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j$$

$$= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \binom{n}{2} x^{n-2} y^2 + \ldots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n$$

<u>Base case</u> $n=1$

$$(x+y)^1 = \binom{1}{0} x^1 + \binom{1}{1} y^1$$

$$x + y = x + y \checkmark$$

<u>Inductive step</u> assume following inductive hypothesis is true:

$\boxed{A}$ $$(x+y)^k = \binom{k}{0} x^k + \binom{k}{1} x^{k-1} y + \binom{k}{2} x^{k-2} y^2 + \ldots + \binom{k}{k-1} x y^{k-1} + \binom{k}{k} y^k$$

- Prove that the thm holds with $(k+1)$:

$$\binom{k+1}{0} x^{k+1} + \binom{k+1}{1} x^k y + \binom{k+1}{2} x^{k-1} y^2 + \ldots + \binom{k+1}{k} x y^k + \binom{k+1}{k+1} y^{k+1}$$

- start by multiplying each side by $(x+y)$

$$(x+y)^k (x+y) = (x+y)^{k+1}$$

$$(x+y)^{k+1} = \left[\binom{k}{0} x^k + \binom{k}{1} x^{k-1} y + \binom{k}{2} x^{k-2} y^2 + \ldots + \binom{k}{k-1} x y^{k-1} + \binom{k}{k} y^k \right] \cdot (x+y)$$

CONT →

4 CONT | multiply $(x+y)$ into system

$$= x\left[\binom{k}{0}x^k + \binom{k}{1}x^{k-1}y + \binom{k}{2}x^{k-2}y^2 + \dots + \binom{k}{k-1}xy^{k-1} + \binom{k}{k}y^k\right]$$

$$+$$

$$y\left[\binom{k}{0}x^k + \binom{k}{1}x^{k-1}y + \binom{k}{2}x^{k-2}y^2 + \dots + \binom{k}{k-1}xy^{k-1} + \binom{k}{k}y^k\right]$$

$$= \left[\binom{k}{0}x^{k+1} + \binom{k}{1}x^k y + \binom{k}{2}x^{k-1}y^2 + \dots + \binom{k}{k-1}x^2y^{k-1} + \binom{k}{k}xy^k\right]$$

A B C

$$+$$

$$\left[\binom{k}{0}x^k y + \binom{k}{1}x^{k-1}y^2 + \binom{k}{2}x^{k-2}y^3 + \dots + \binom{k}{k-1}xy^k + \binom{k}{k}y^{k+1}\right]$$

- using pascals identity we can find and combine like terms

$$= \binom{k}{0}x^{k+1} + \boxed{\left[\binom{k}{1}+\binom{k}{0}\right]x^k y}^{\boxed{A}} + \boxed{\left[\binom{k}{2}+\binom{k}{1}\right]}^{\boxed{B}}x^{k-1}y^2 + \binom{k}{2}x^{k-2}y^3 +$$

$$\binom{k}{k-1}x^2y^{k-1} + \boxed{\left[\binom{k}{k}+\binom{k}{k-1}\right]}^{\boxed{C}}xy^k + \binom{k}{k}y^{k+1}$$

- the result of $\boxed{A}$ $\boxed{B}$ and $\boxed{C}$ are as follows:

$\boxed{A}$ $\binom{k}{1}+\binom{k}{0}x^k y = \binom{k+1}{1}x^k y$

$\boxed{B}$ $\binom{k}{2}+\binom{k}{1}x^{k-1}y^2 = \binom{k+1}{2}x^{k-1}y^2$

$\boxed{C}$ $\binom{k}{k}+\binom{k}{k-1}xy^k = \binom{k+1}{k}xy^k$

- substituting these back into the equation

CONT →

4 CONT

$$= \binom{k}{0} x^{k+1} + \binom{k+1}{1} x^{k} y + \binom{k+1}{2} x^{k-1} y^{2} + \underline{\underline{\binom{k}{2} x^{k-2} y^{3}}} +$$

$$\underline{\underline{\binom{k}{k-1} x^{2} y^{k-1}}} + \binom{k+1}{k} x y^{k} + \binom{k}{k} y^{k+1}$$

the two portions of the eq. that remain in "k" would have a "partner" if the eq. was fully expanded for example (using #2 of this HW)

let $k = 5$

$$\binom{5}{0} x^{5} y^{0} + \binom{5}{1} x^{4} y + \binom{5}{2} x^{3} y^{2} + \binom{5}{3} x^{2} y^{3} +$$

$$\binom{5}{4} x y^{4} + \binom{5}{5} y^{5}$$

let $k = k+1 \therefore k = 6 \rightarrow$ multiply by $(x+y)$

$$= x\left[\binom{5}{0} x^{5} + \binom{5}{1} x^{4} y + \binom{5}{2} x^{3} y^{2} + \binom{5}{3} x^{2} y^{3} + \binom{5}{4} x y^{4} + \binom{5}{5} y^{5}\right]$$

$$+$$

$$y\left[\binom{5}{0} x^{5} + \binom{5}{1} x^{4} y + \binom{5}{2} x^{3} y^{2} + \binom{5}{3} x^{2} y^{3} + \binom{5}{4} x y^{4} + \binom{5}{5} y^{5}\right]$$

$$= \underbrace{\binom{5}{0} x^{6}}_{\boxed{A}} + \underbrace{\binom{5}{1} x^{5} y}_{\boxed{B}} + \underbrace{\binom{5}{2} x^{4} y^{2}}_{\boxed{C}} + \underbrace{\binom{5}{3} x^{3} y^{3}}_{\boxed{D}} + \underbrace{\binom{5}{4} x^{2} y^{4}}_{\boxed{E}} + \binom{5}{5} x y^{5}$$

$$+ \binom{5}{0} x^{5} y + \binom{5}{1} x^{4} y^{2} + \underline{\binom{5}{2} x^{3} y^{3}} + \underline{\binom{5}{3} x^{2} y^{4}} + \underline{\binom{5}{4} x y^{5}} + \underline{\binom{5}{5} y^{6}}$$

$$= \binom{5}{0} x^{6} + \underbrace{\binom{6}{1}}_{\boxed{A}} x^{5} y + \underbrace{\binom{6}{2}}_{\boxed{B}} x^{4} y^{2} + \underbrace{\binom{6}{3}}_{\boxed{C}} x^{3} y^{3} + \underbrace{\binom{6}{4}}_{\boxed{D}} x^{2} y^{4} + \underbrace{\binom{6}{5}}_{\boxed{E}} x y^{5} + \binom{5}{5} y^{6}$$

All "middles" get combined!

CONT →

4 CONT

- Now that we know all "midales" combine we are left with

$$\underbrace{\binom{k}{0}x^{k+1}}_{\frac{k!}{0!\,k-0!} = \frac{k!}{1(k!)} = 1} + \binom{k+1}{1}x^{k}y + \binom{k+1}{2}x^{k-1}y^{2} + \dots + \binom{k+1}{k}xy^{k} + \underbrace{\binom{k}{k}y^{k+1}}_{\frac{k!}{k!(k-k)!} = 1}$$

- both the left side $\binom{k}{0}x^{k+1}$ & right side $\binom{k}{k}y^{k+1}$ go to 1 so we are left w/

$$\binom{k+1}{1}x^{k}y + \binom{k+1}{2}x^{k-1}y^{2} + \dots + \binom{k+1}{k}xy^{k}$$

which is equal to our induction hypothesis!